

Shapes Areas

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

You are given an integer t that determines the type of shape:

- $t = 1$: Rectangle (given values a, b as doubles)
- $t = 2$: Square (given value a as double)
- $t = 3$: Circle (given value r as double)

Your task is to compute the area of the given shape.

Formulas:

$$\text{Rectangle area} = a \times b$$

$$\text{Square area} = a^2$$

$$\text{Circle area} = \pi r^2$$

- Use $\pi = 3.14159$.
- **print only 2 decimal places of the final answer** (cout « fixed « setprecision(2) « yourvariable;)
- **add the following to your file header:** <iomanip>

Input

The first line contains an integer t .

If $t = 1$, the next line contains two values a, b ($1 \leq a, b \leq 100$).

If $t = 2$, the next line contains one value a ($1 \leq a \leq 100$).

If $t = 3$, the next line contains one value r ($1 \leq r \leq 100$).

Output

Print the area of the given shape.

Examples

standard input	standard output
1 2.3 3.3	7.59
3 2.5	19.63